

Linear Regression Analysis for The spermidine acetyltransferase SpeG regulates transcription of the small RNA rprA

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Analysis performed for:

The spermidine acetyltransferase SpeG regulates transcription of the small RNA RprA

Linda I. Hu, Ekaterina V. Filippova, Joseph Dang, Sergii Pshenychnyi, Jiapeng Ruan, OlgaKiryukhina, Wayne F. Anderson, Misty L. Kuhn, Alan J. Wolfe

bioRxiv 462937; doi: <https://doi.org/10.1101/462937>

Methods Excerpt From Paper

“To determine whether experimental results were statistically significant, a linear regression was performed, comparing all experimental groups with their respective vector controls. All of the regressions used were set up as follows: the calculated rprApromoter activity was the response variable, the overexpressed plasmids or mutant were the explanatory variable, and time was a random effect. OD was not included as an effect on activity as it is already used in the calculation of activity. Time as a random effect was chosen based on the question asked: Accounting for the effects of time on activity does the experimental group in question significantly affect overall rprApromoter activity? The significance threshold was set at 0.05. The open source program R (version 3.3.2) and packages “lmerTest”, “ggplot2”, and “moments” were used to visualize and analyze the data (76,77,78,79).”

Activity = 1 Miller Unit = 1000 x (Abs420 - (1.75 x Abs550)) / (T x V x Abs600) We are then able to disregard OD

Hypothesis: Treatment groups (plasmids or mutant) have an effect on rprA Promoter activity compared to their respected vector controls.

Note: Although we check for normality of the data during analysis in order to determine the correct test to use for the question asked for the final test selected normal data is not necessary.

Analysis Details

Loading Libraries

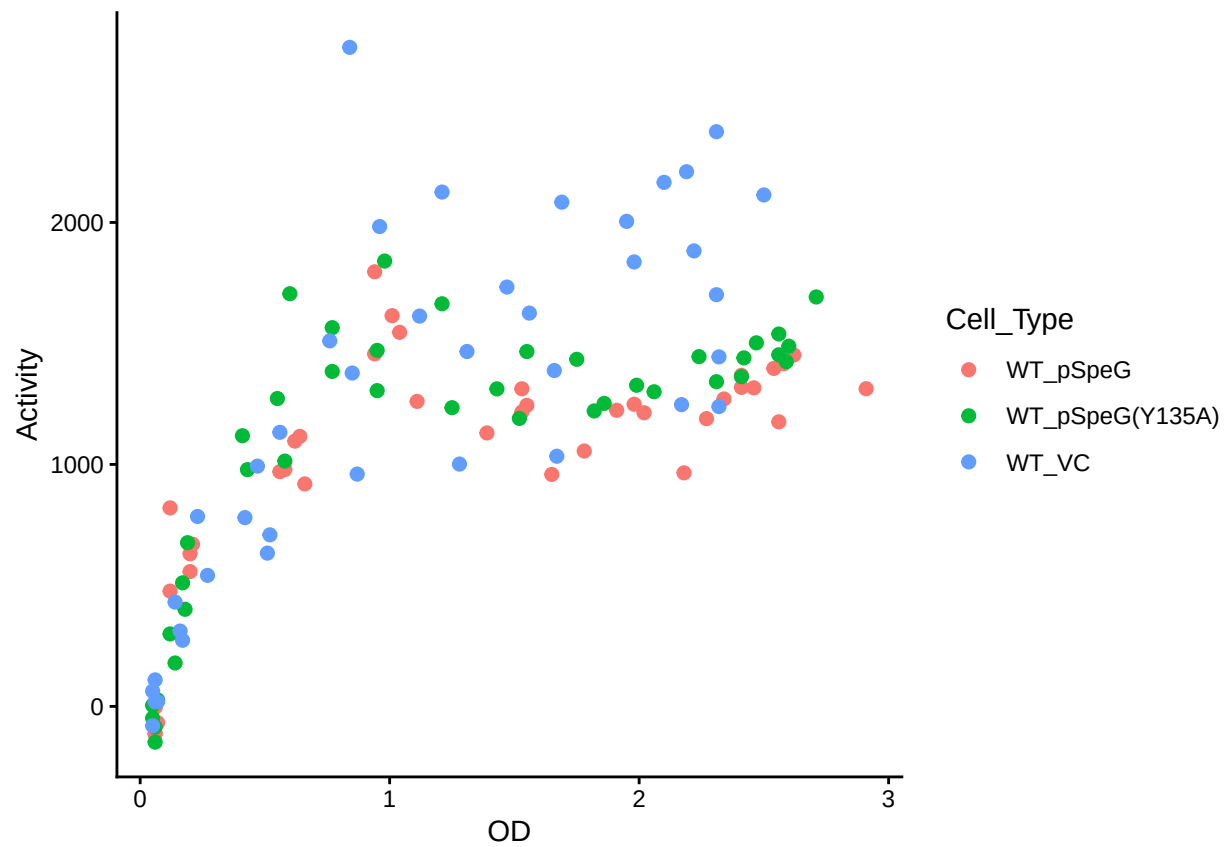
```
library(lmerTest)
library(ggplot2)
library(moments)
```

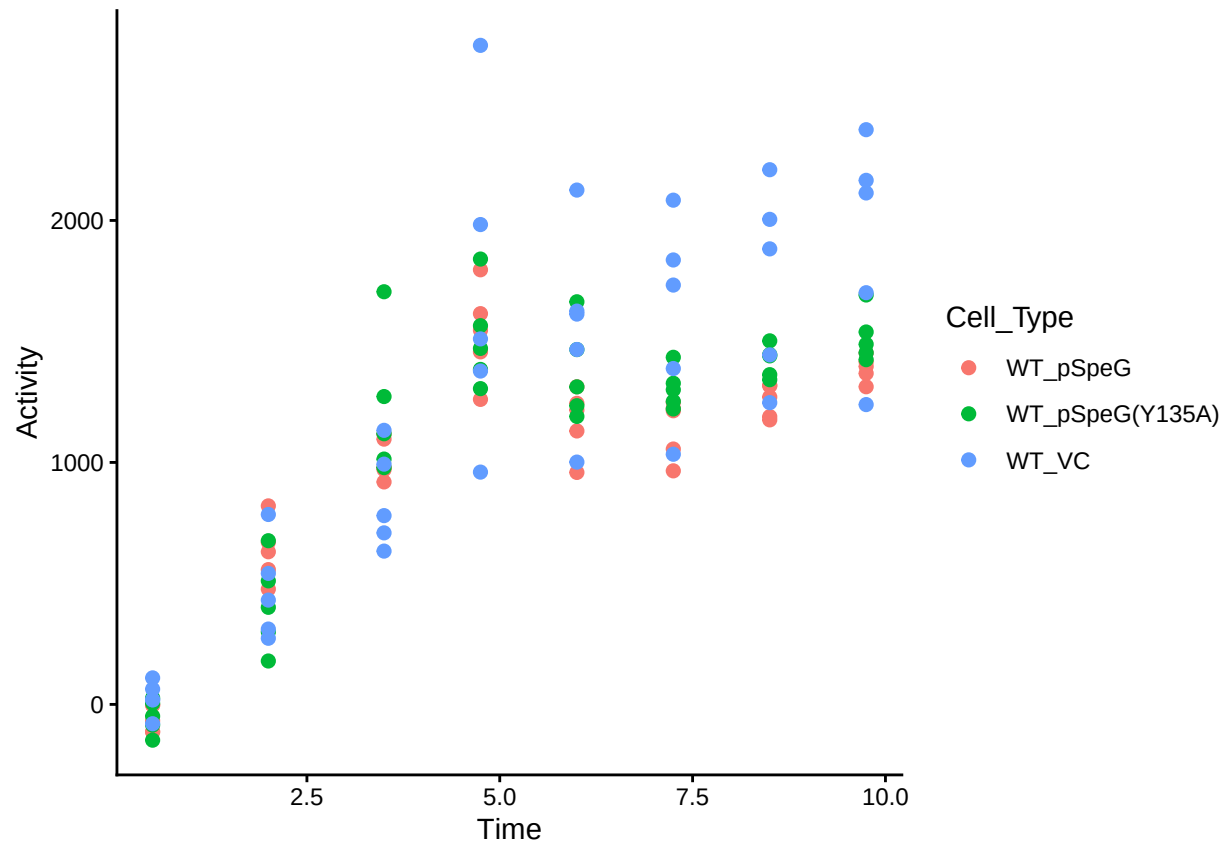
Figure 3A

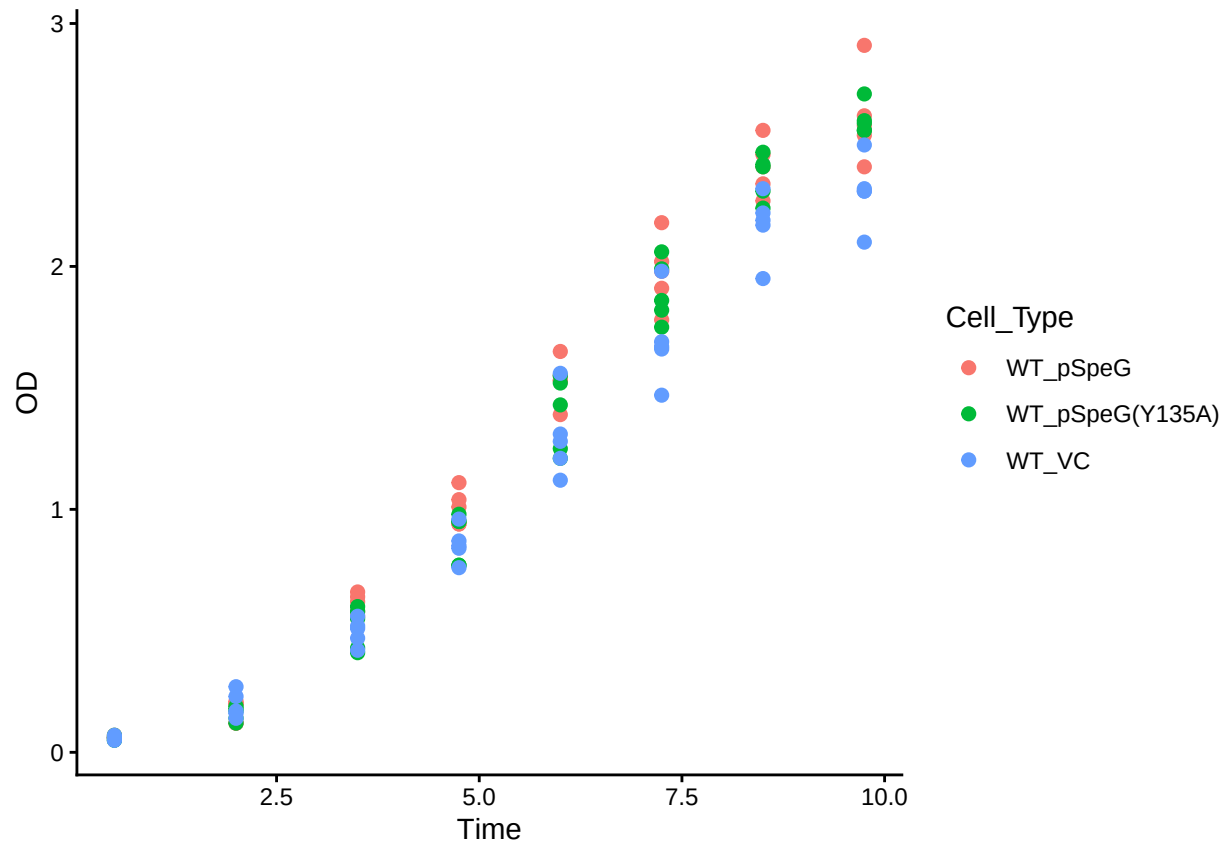
Taking a look at the data

##	Time	Cell_Type	Activity	OD	AcT
## 1	0.50	WT_pSpeG	-113.20	0.06	-6.7920
## 2	2.00	WT_pSpeG	557.07	0.20	111.4140
## 3	3.50	WT_pSpeG	978.92	0.58	567.7736
## 4	4.75	WT_pSpeG	1796.11	0.94	1688.3434
## 5	6.00	WT_pSpeG	1312.65	1.53	2008.3545
## 6	7.25	WT_pSpeG	1223.74	1.91	2337.3434
## 7	8.50	WT_pSpeG	1318.49	2.41	3177.5609
## 8	9.75	WT_pSpeG	1416.21	2.58	3653.8218
## 9	0.50	WT_pSpeG	-2.93	0.06	-0.1758
## 10	2.00	WT_pSpeG	630.85	0.20	126.1700

##	Time	Cell_Type	Activity	OD	AcT
## 111	8.50	WT_VC	2209.6981	2.19	4839.238806
## 112	9.75	WT_VC	2375.1228	2.31	5486.533763
## 113	0.50	WT_VC	19.2749	0.07	1.349243
## 114	2.00	WT_VC	541.4088	0.27	146.180370
## 115	3.50	WT_VC	1132.6605	0.56	634.289879
## 116	4.75	WT_VC	1983.0903	0.96	1903.766724
## 117	6.00	WT_VC	1625.3814	1.56	2535.595050
## 118	7.25	WT_VC	1836.6935	1.98	3636.653162
## 119	8.50	WT_VC	1882.5694	2.22	4179.303988
## 120	9.75	WT_VC	2113.5969	2.50	5283.992243



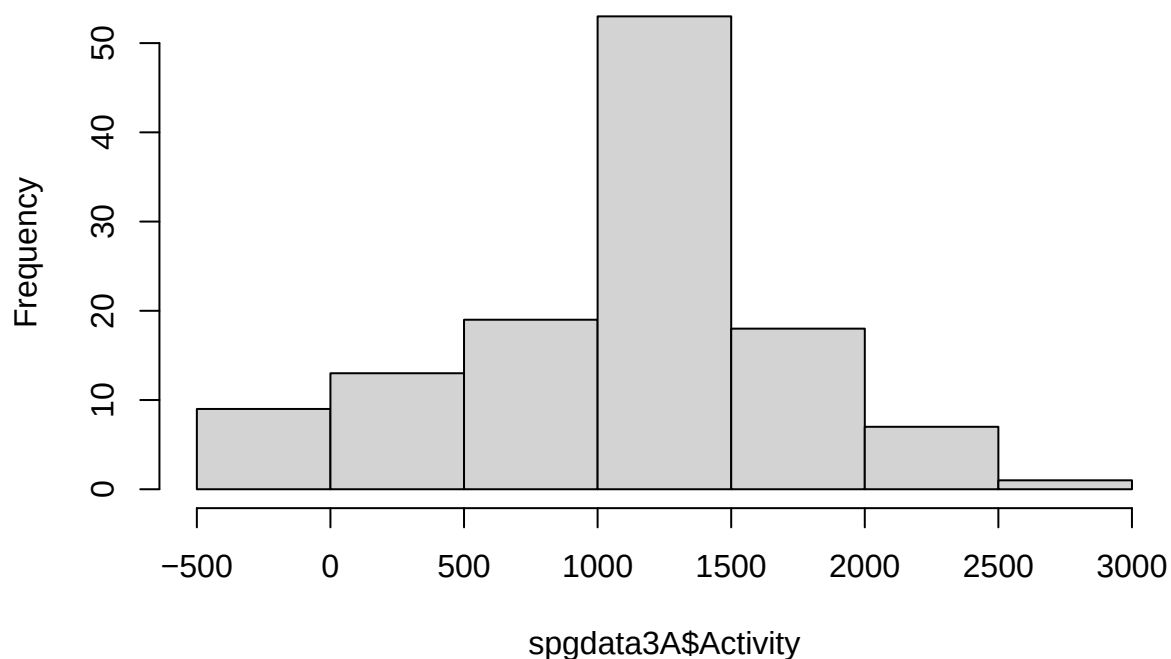




Checking data “Activity” for normality

```
hist(spgdata3A$Activity)
```

Histogram of spgdata3A\$Activity



```
skewness(spgdata3A$Activity)
```

```
## [1] -0.3586738
```

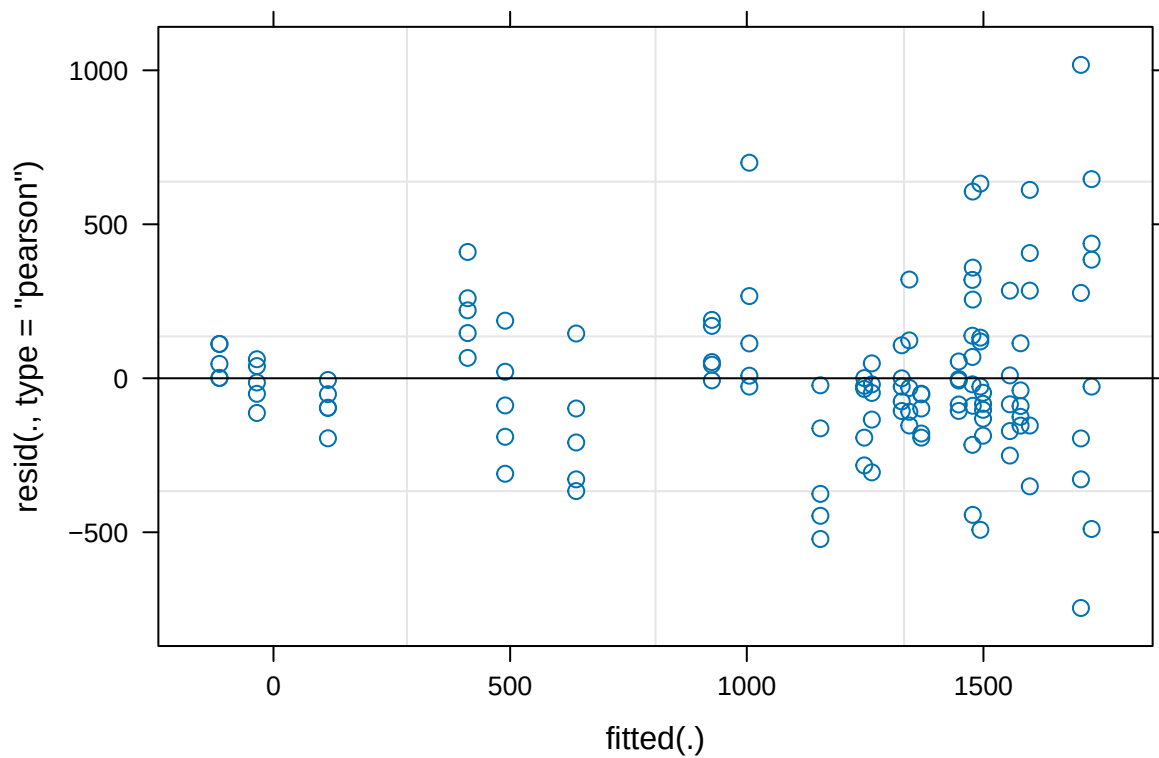
```
kurtosis(spgdata3A$Activity)
```

```
## [1] 2.775568
```

Accounting for time as a random effect

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [  
## lmerModLmerTest]  
## Formula: Activity ~ Cell_Type + (1 | Time)  
## Data: spgdata3A  
##  
## REML criterion at convergence: 1685.6  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max   
## -2.7291 -0.4823 -0.0985  0.4143  3.7234   
##  
## Random effects:  
## Groups   Name                Variance Std.Dev.  
## Time     (Intercept) 338869    582.1
```

```
## Residual          74707   273.3
## Number of obs: 120, groups: Time, 8
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    1239.190    210.301    7.411   5.892 0.000488 ***
## Cell_TypeSpgvsCtrl -229.283     61.117  110.000  -3.752 0.000282 ***
## Cell_TypeY135AvsCtrl -150.090     61.117  110.000  -2.456 0.015623 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) Cl_TSC
## Cl1_TypSpgC -0.145
## Cl1_TY135AC -0.145  0.500
```



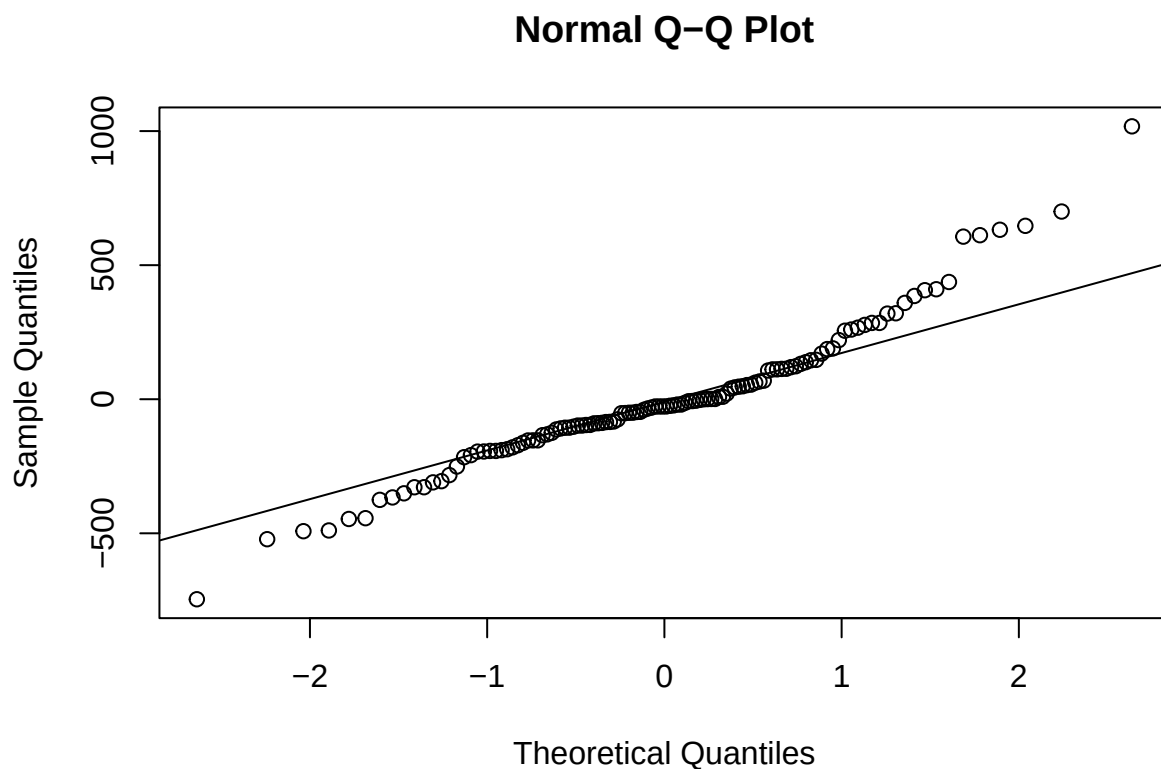
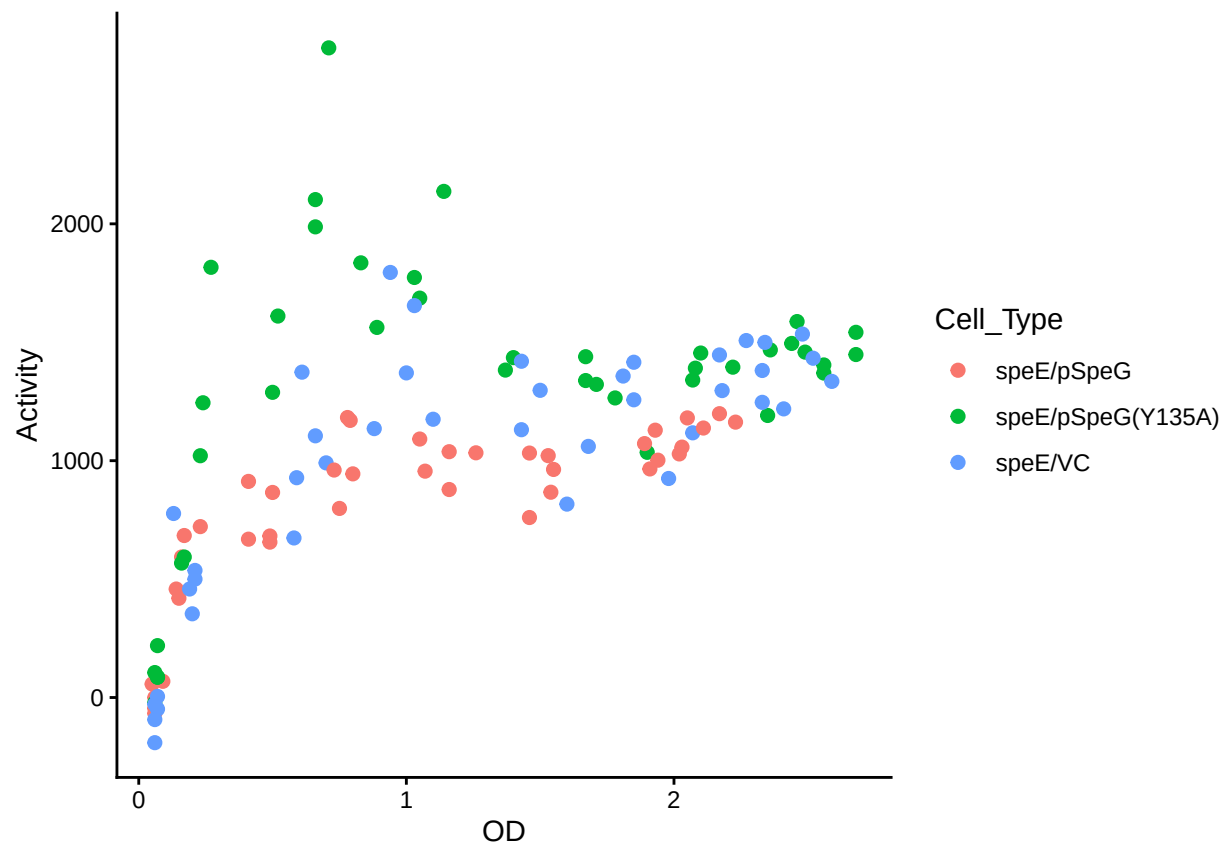


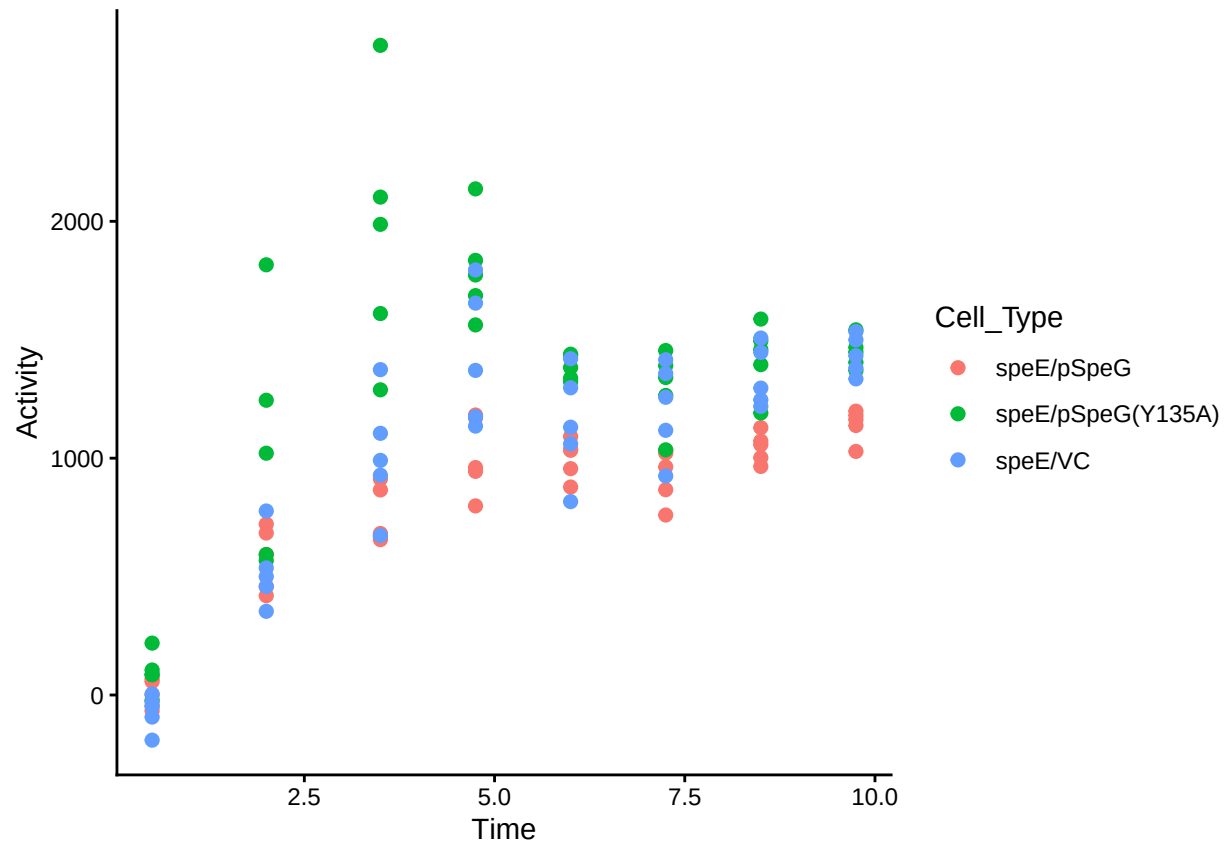
Figure 3B

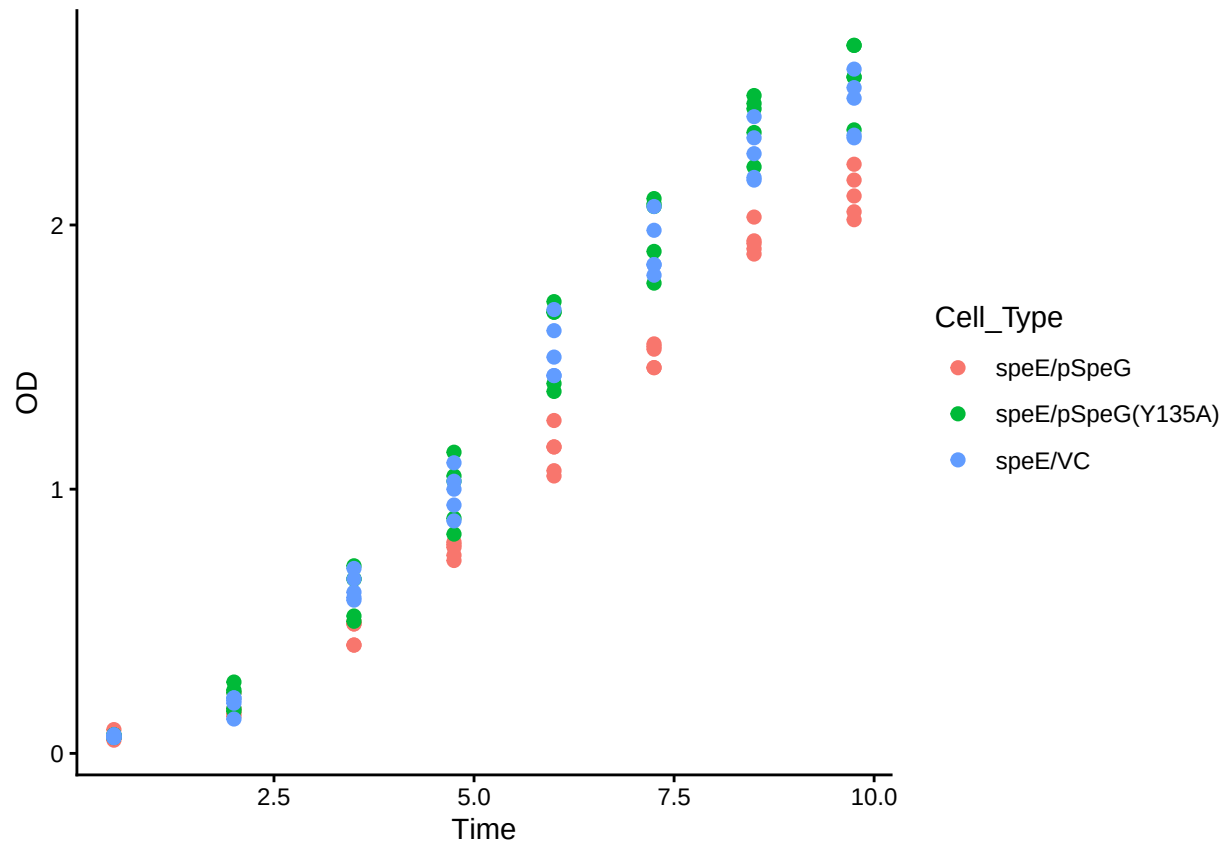
Take a look at the data

##	X	Time	Cell_Type	variable	Activity	OD
## 1	1	0.5	speE/pSpeG	A	0.00	0.06
## 2	2	0.5	speE/pSpeG	B	-45.63	0.06
## 3	3	0.5	speE/pSpeG	C	68.44	0.09
## 4	4	0.5	speE/pSpeG	D	-67.07	0.06
## 5	5	0.5	speE/pSpeG	E	57.03	0.05
## 6	6	0.5	speE/pSpeG(Y135A)	A	219.18	0.07
## 7	7	0.5	speE/pSpeG(Y135A)	B	105.46	0.06
## 8	8	0.5	speE/pSpeG(Y135A)	C	-21.92	0.06
## 9	9	0.5	speE/pSpeG(Y135A)	D	85.99	0.07
## 10	10	0.5	speE/pSpeG(Y135A)	E	85.99	0.07
##	X	Time	Cell_Type	variable	Activity	OD
## 111	111	9.75	speE/pSpeG(Y135A)	A	1542.00	2.68
## 112	112	9.75	speE/pSpeG(Y135A)	B	1404.23	2.56
## 113	113	9.75	speE/pSpeG(Y135A)	C	1467.73	2.36
## 114	114	9.75	speE/pSpeG(Y135A)	D	1370.11	2.56
## 115	115	9.75	speE/pSpeG(Y135A)	E	1448.21	2.68
## 116	116	9.75	speE/VC	A	1499.51	2.34
## 117	117	9.75	speE/VC	B	1380.84	2.33
## 118	118	9.75	speE/VC	C	1334.80	2.59

##	119	119	9.75	speE/VC	D	1534.60	2.48
##	120	120	9.75	speE/VC	E	1432.60	2.52



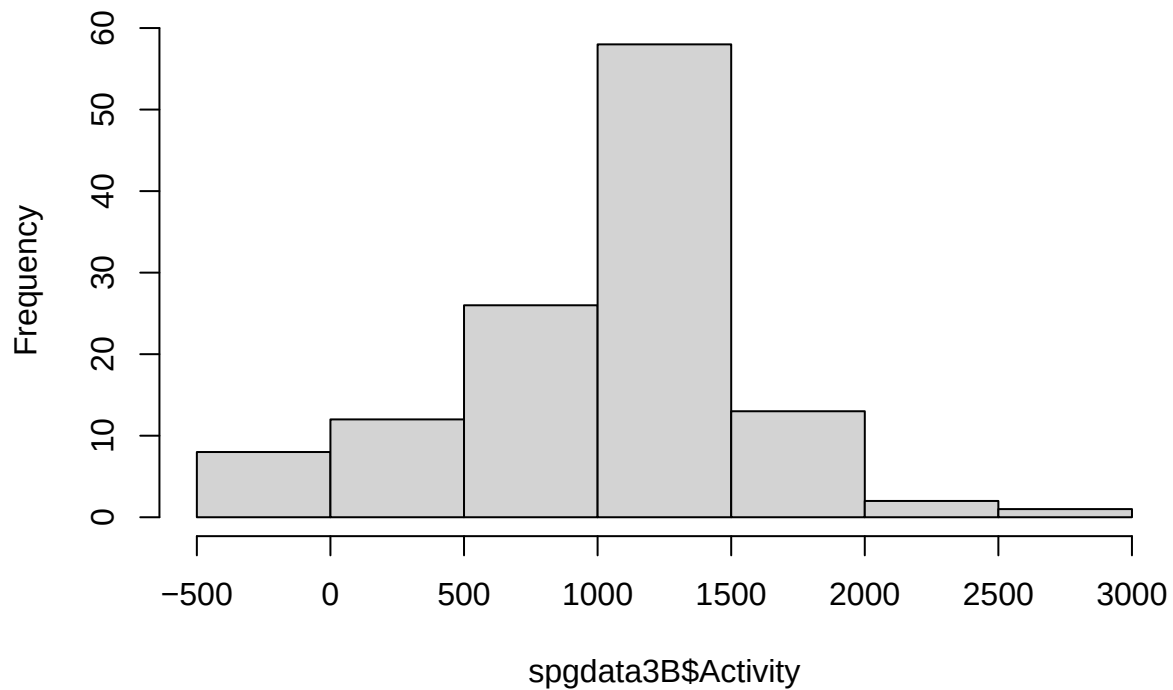




Checking data “Activity” for normality

```
hist(spgdata3B$Activity)
```

Histogram of spgdata3B\$Activity



```
skewness(spgdata3B$Activity)
```

```
## [1] -0.3075729
```

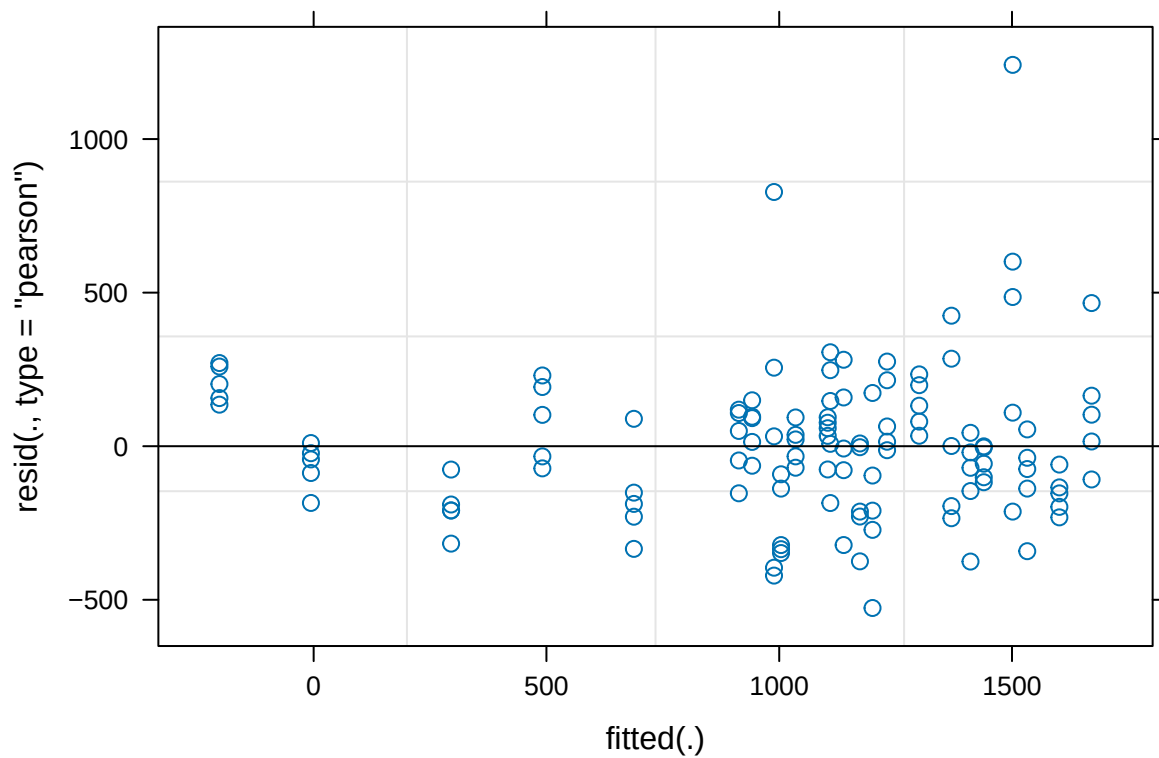
```
kurtosis(spgdata3B$Activity)
```

```
## [1] 3.293223
```

Accounting for time as a random effect

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [  
## lmerModLmerTest]  
## Formula: Activity ~ Cell_Type + (1 | Time)  
## Data: spgdata3B  
##  
## REML criterion at convergence: 1665.7  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max   
## -2.0794 -0.5987 -0.0234  0.4275  4.9015   
##  
## Random effects:  
## Groups   Name                Variance Std.Dev.  
## Time     (Intercept) 212956    461.5
```

```
## Residual          64164   253.3
## Number of obs: 120, groups: Time, 8
##
## Fixed effects:
##              Estimate Std. Error      df t value    Pr(>|t|)
## (Intercept)    1004.138    167.999    7.561   5.977  0.000413 ***
## Cell_TypeSpgvsCtrl -196.538     56.641  110.000  -3.470  0.000745 ***
## Cell_TypeY135AvsCtrl 301.037     56.641  110.000   5.315 0.000000564 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) Cl_TSC
## Cl1_TypSpgC -0.169
## Cl1_TY135AC -0.169  0.500
```



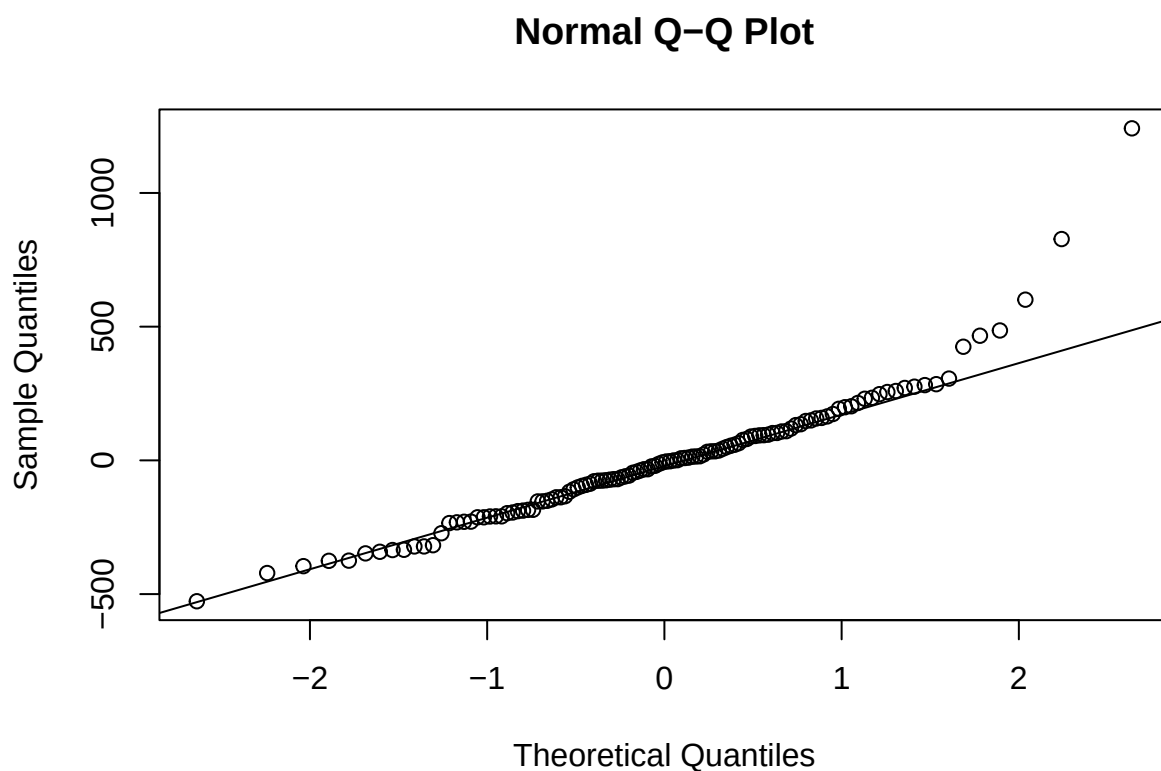


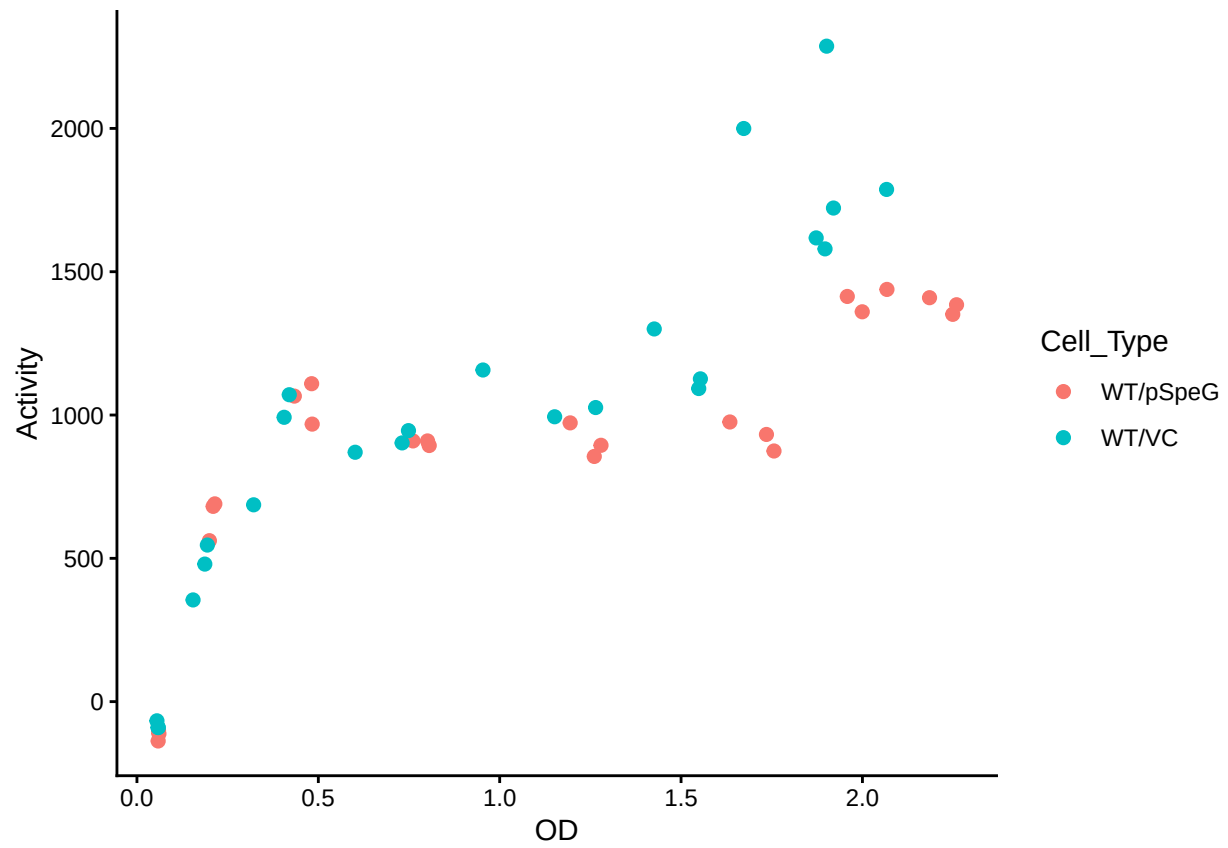
Figure 1A

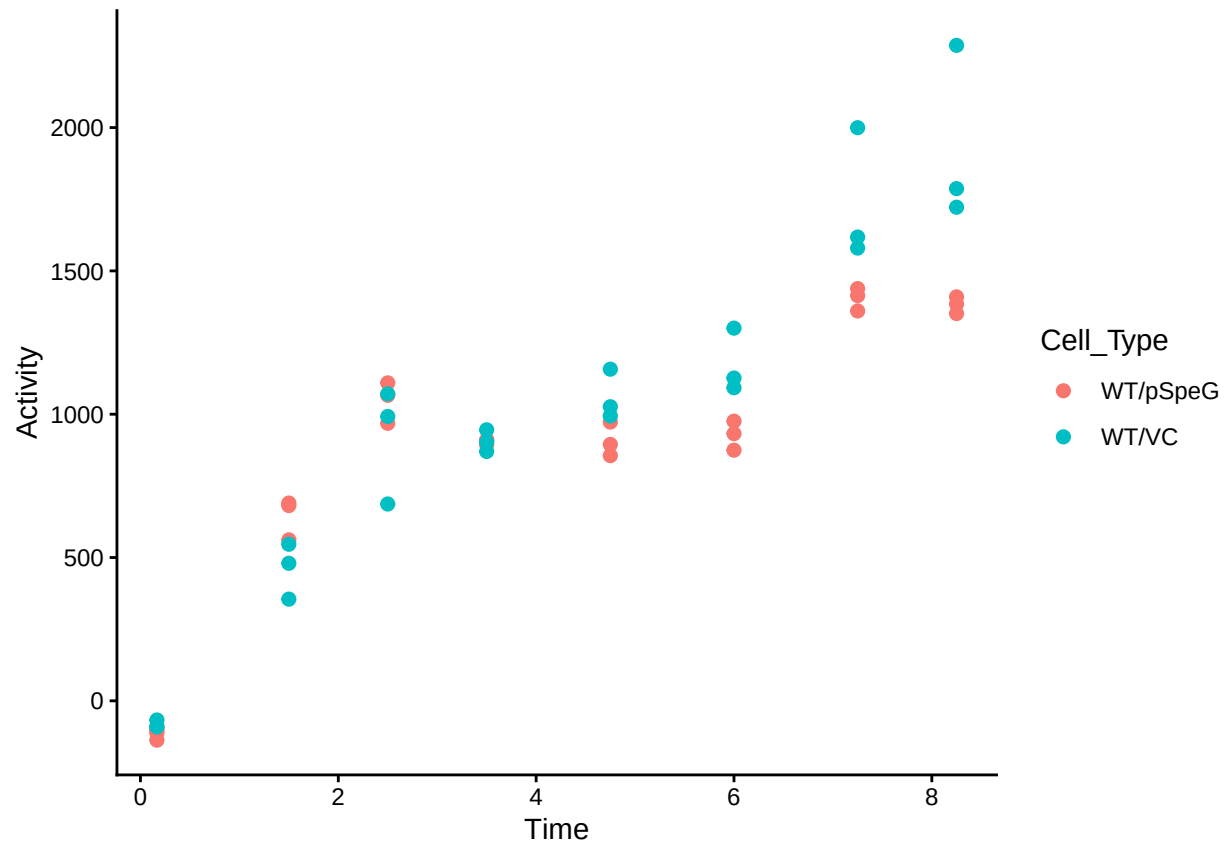
Take a look at the data

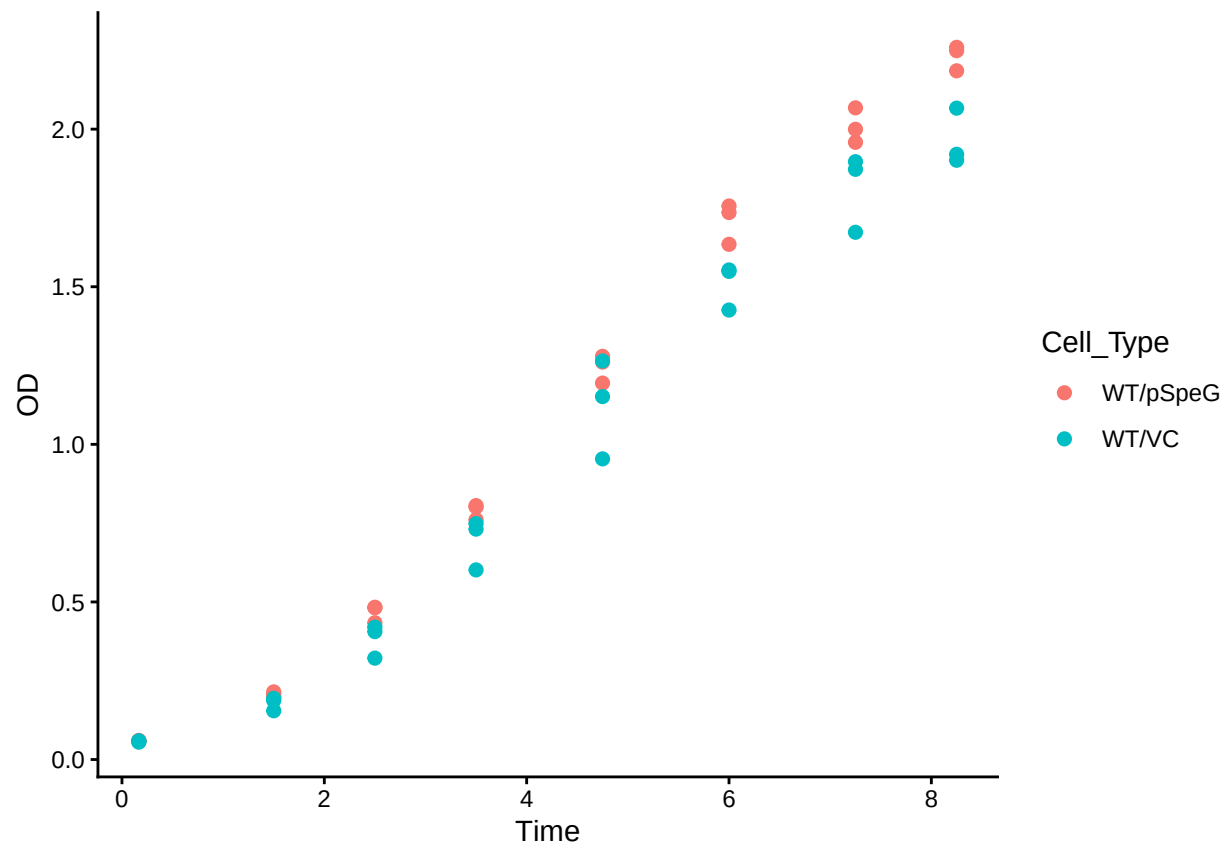
##	Time	Cell_Type	Activity	OD
## 1	0.1666667	WT/pSpeG	-137.3	0.0585
## 2	0.1666667	WT/pSpeG	-101.3	0.0595
## 3	0.1666667	WT/pSpeG	-112.3	0.0606
## 4	1.5000000	WT/pSpeG	561.6	0.1997
## 5	1.5000000	WT/pSpeG	681.3	0.2101
## 6	1.5000000	WT/pSpeG	690.0	0.2147
## 7	2.5000000	WT/pSpeG	1066.0	0.4339
## 8	2.5000000	WT/pSpeG	1109.3	0.4816
## 9	2.5000000	WT/pSpeG	968.3	0.4831
## 10	3.5000000	WT/pSpeG	909.6	0.7610

##	Time	Cell_Type	Activity	OD
## 39	4.75	WT/VC	1026.1	1.2645
## 40	6.00	WT/VC	1300.3	1.4261
## 41	6.00	WT/VC	1092.5	1.5487
## 42	6.00	WT/VC	1126.2	1.5533
## 43	7.25	WT/VC	1999.7	1.6728
## 44	7.25	WT/VC	1618.0	1.8725
## 45	7.25	WT/VC	1579.7	1.8969
## 46	8.25	WT/VC	2287.0	1.9013

47 8.25 WT/VC 1722.2 1.9202
48 8.25 WT/VC 1787.1 2.0666



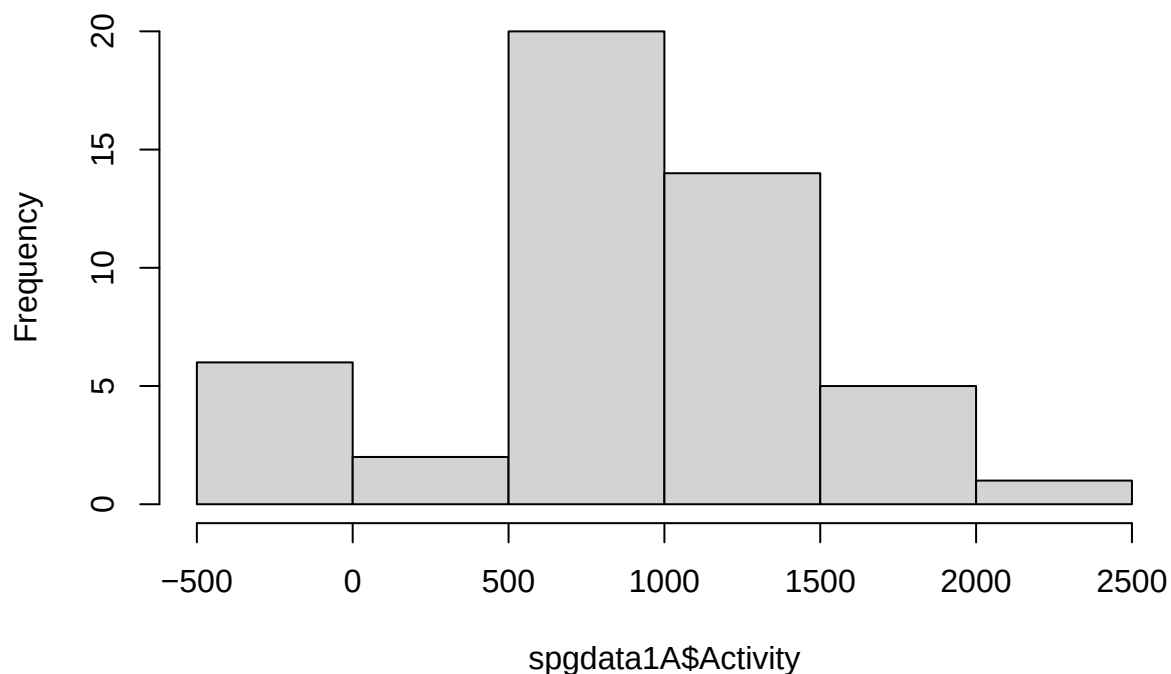




Checking data “Activity” for normality

```
hist(spgdata1A$Activity)
```

Histogram of spgdata1A\$Activity



```
skewness(spgdata1A$Activity)
```

```
## [1] -0.2193354
```

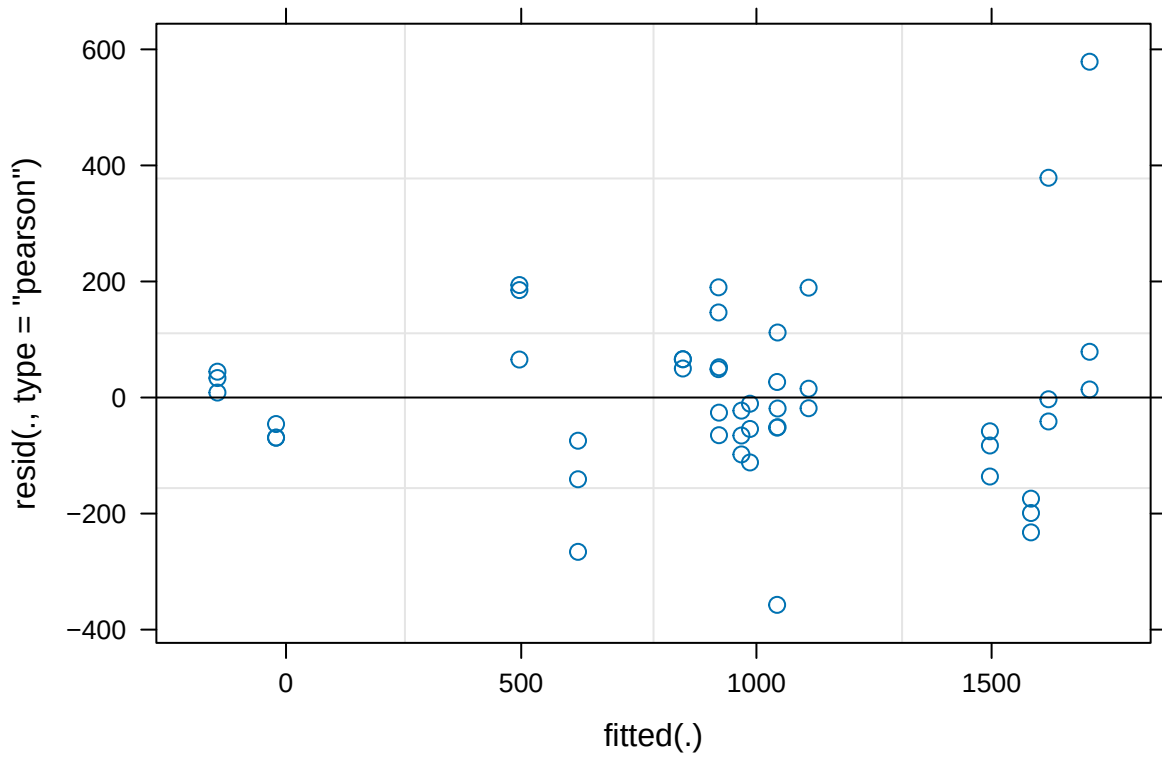
```
kurtosis(spgdata1A$Activity)
```

```
## [1] 3.175003
```

Accounting for time as a random effect

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [  
## lmerModLmerTest]  
## Formula: Activity ~ Cell_Type + (1 | Time)  
## Data: spgdata1A  
##  
## REML criterion at convergence: 638.1  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max   
## -2.1131 -0.4091 -0.1107  0.3281  3.4225   
##  
## Random effects:  
## Groups   Name                Variance Std.Dev.  
## Time     (Intercept) 303288    550.7
```

```
## Residual                28589   169.1
## Number of obs: 48, groups: Time, 8
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept) 1012.129   197.743    7.218   5.118  0.00125 **
## Cell_Type1  -124.596    48.810   39.000  -2.553  0.01472 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## Cell_Type1 -0.123
```



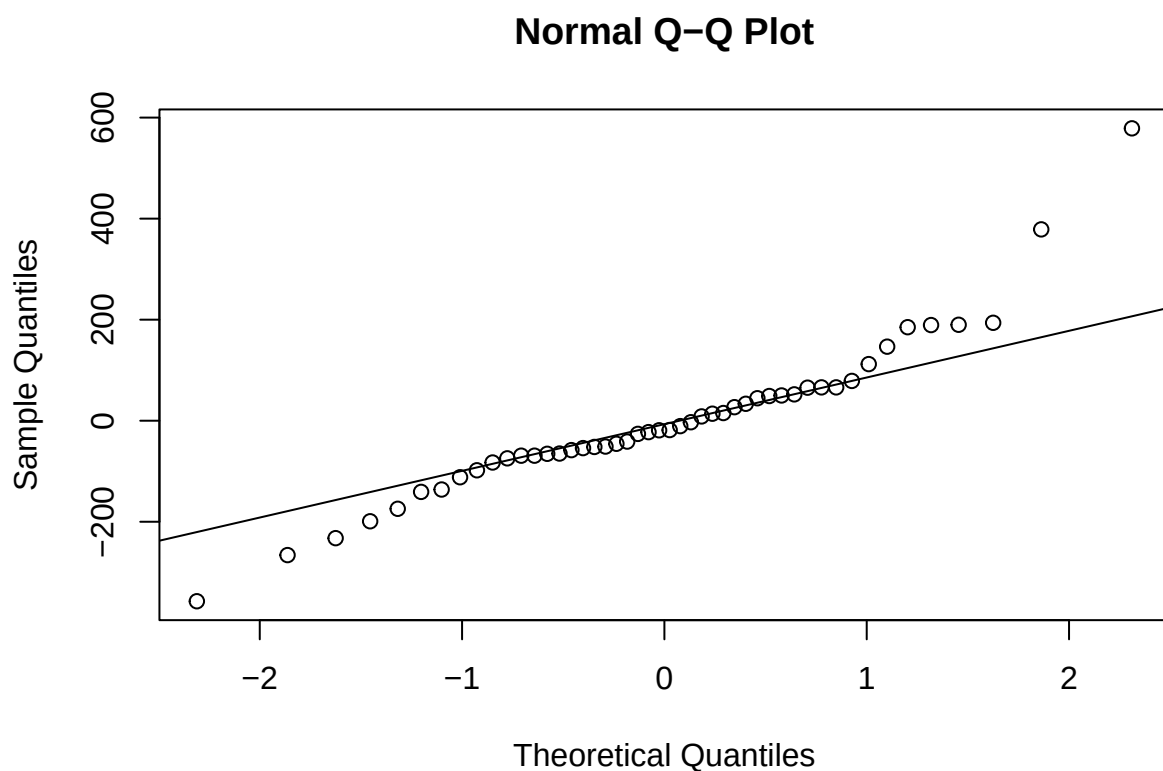


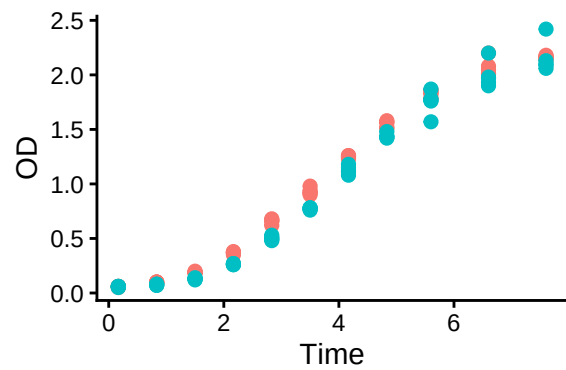
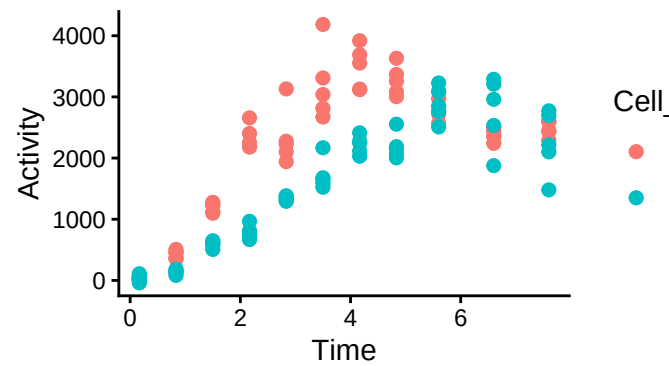
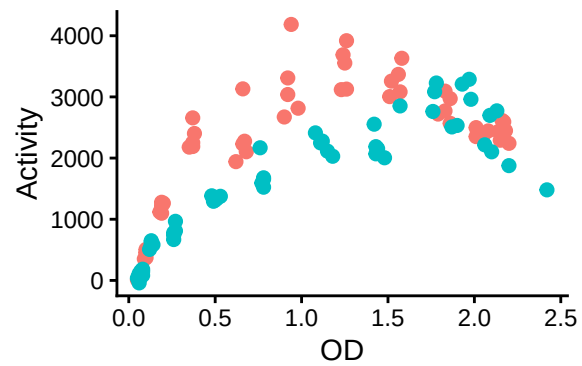
Figure 1B

Take a look at the data

##	Time	Cell_Type	Activity	OD
## 1	0.1666667	speG	57.38	0.06
## 2	0.1666667	speG	10.99	0.06
## 3	0.1666667	speG	10.99	0.06
## 4	0.1666667	speG	36.10	0.06
## 5	0.1666667	speG	10.99	0.05
## 6	0.8333333	speG	507.06	0.10
## 7	0.8333333	speG	371.76	0.10
## 8	0.8333333	speG	452.81	0.10
## 9	0.8333333	speG	457.04	0.10
## 10	0.8333333	speG	351.85	0.09

##	Time	Cell_Type	Activity	OD
## 101	6.6	WT	3287.90	1.97
## 102	6.6	WT	2960.83	1.98
## 103	6.6	WT	3209.08	1.93
## 104	6.6	WT	2533.59	1.90
## 105	6.6	WT	1876.65	2.20
## 106	7.6	WT	2698.11	2.09
## 107	7.6	WT	2772.09	2.13
## 108	7.6	WT	2216.92	2.06

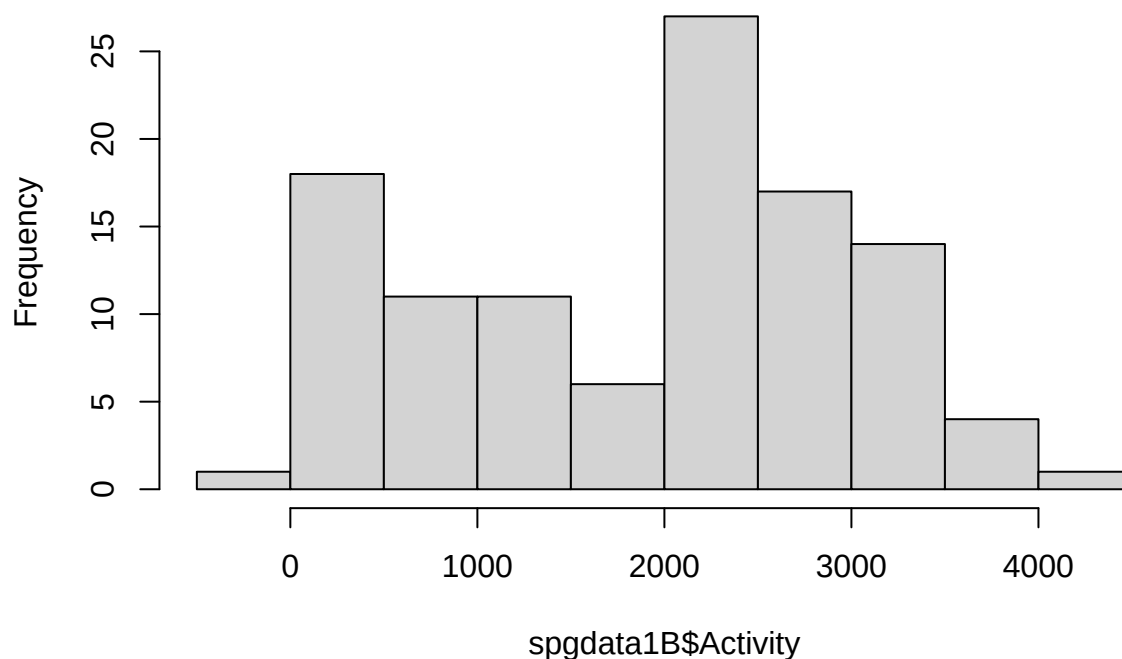
```
## 109 7.6 WT 2102.73 2.10
## 110 7.6 WT 1480.42 2.42
```



Checking data “Activity” for normality

```
hist(spgdata1B$Activity)
```

Histogram of spgdata1B\$Activity



```
skewness(spgdata1B$Activity)
```

```
## [1] -0.2433069
```

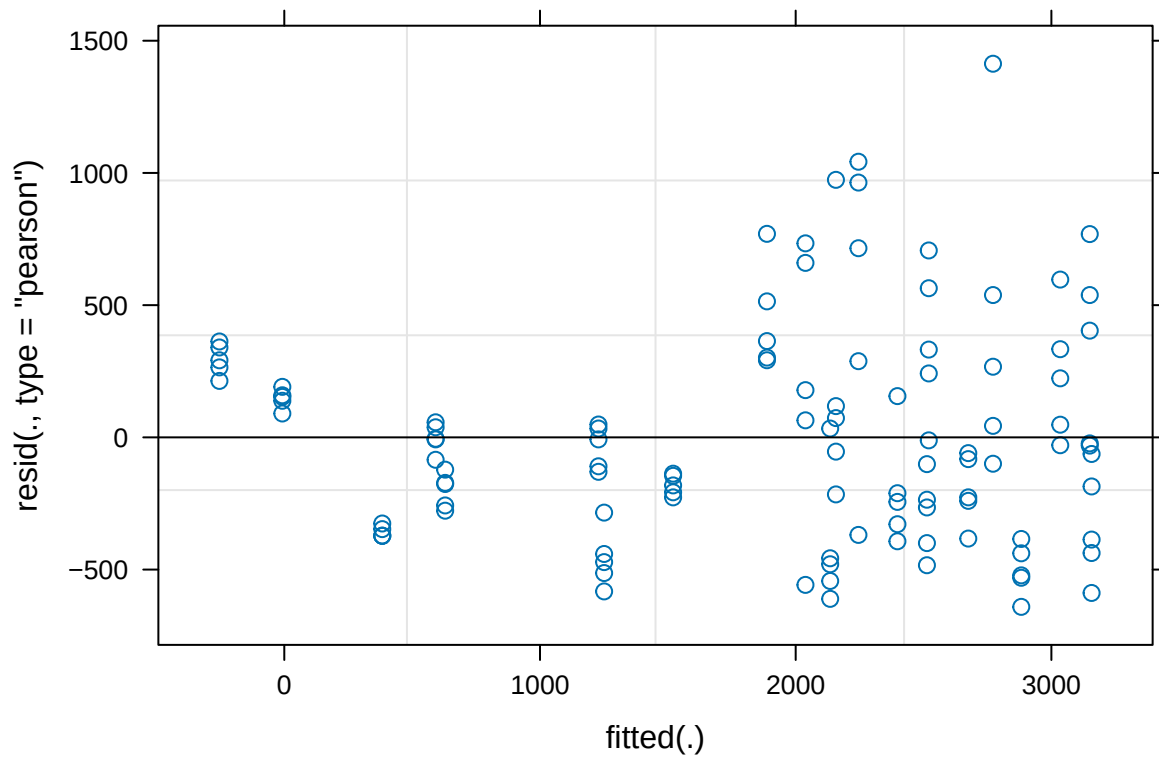
```
kurtosis(spgdata1B$Activity)
```

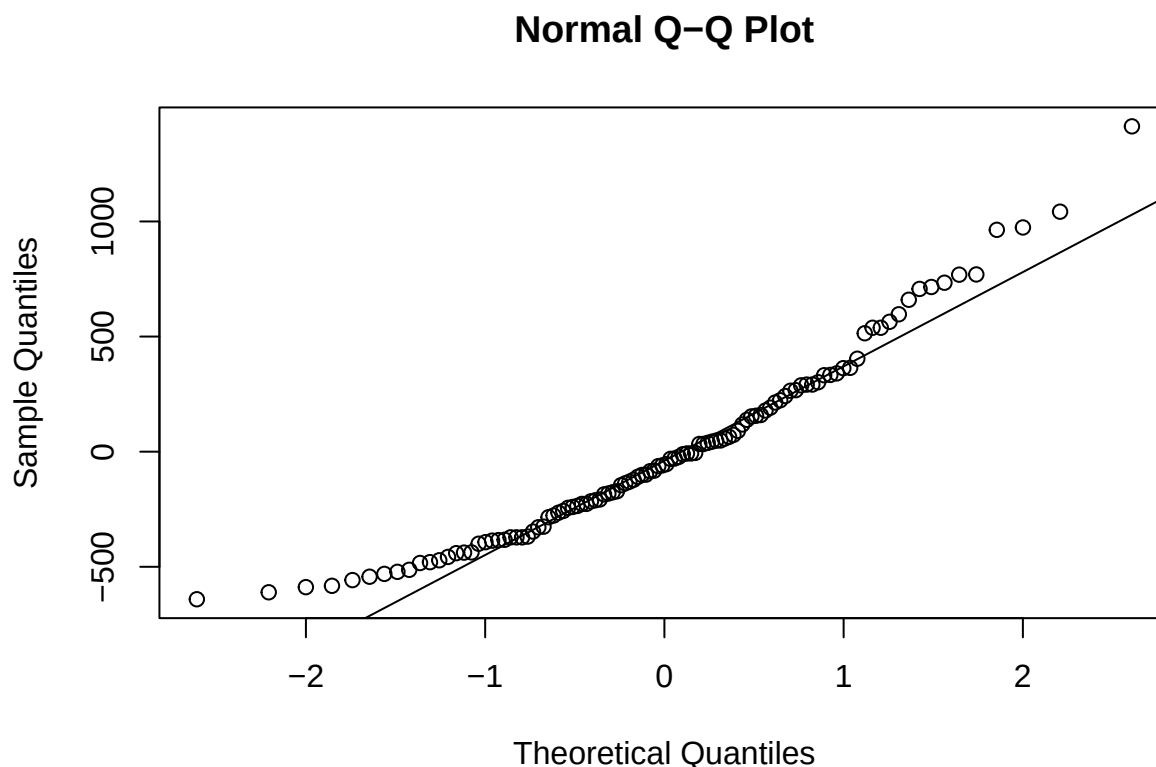
```
## [1] 1.941201
```

Accounting for time as a random effect

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [  
## lmerModLmerTest]  
## Formula: Activity ~ Cell_Type + (1 | Time)  
## Data: spgdata1B  
##  
## REML criterion at convergence: 1665.3  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max   
## -1.4874 -0.7320 -0.1317  0.5503  3.2791   
##  
## Random effects:  
## Groups   Name                Variance Std.Dev.  
## Time     (Intercept) 1052255  1025.8
```

```
## Residual          185699   430.9
## Number of obs: 110, groups: Time, 11
##
## Fixed effects:
##           Estimate Std. Error    df t value      Pr(>|t|)
## (Intercept)  1541.12    314.70   10.35   4.897    0.000566 ***
## Cell_Type1    636.82     82.17   98.00   7.750 0.000000000000865 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## Cell_Type1 -0.131
```





Conclusion Excerpts From Paper

When SpeG was overexpressed from a plasmid in the reference strain, PrprA activity was reduced compared to the vector control during late exponential growth and during the transition into early stationary phase ($OD > 1.0$, Fig 1A, linear regression analysis $t = -2.553$, $p = 0.01472$). When speG was deleted, PrprA activity increased in the isogenic speG mutant compared to its wild-type parent (Fig 1B, linear regression analysis $t = 7.750$, $p = 8.65E-12$). Based on these results, we conclude that SpeG inhibits transcription from PrprA.

...

Fig 3. The effect of overexpressing SpeG or SpeG(Y135A) in WT cells and overexpressing SpeG in the speE mutant on rprA promoter activity. A. WT cells carrying the PrprA-lacZ fusion (AJW3759) were transformed with either pspeG(pCA24n-speG), pspeG(Y135A)(pCA24n-speG(Y135A)), or the VC (pCA24n) and grown in TB7 supplemented with 50 M IPTG and chloramphenicol to maintain the plasmid. Cell growth and β -galactosidase activity were assayed. The values represent average promoter activity with standard deviations of five independent cultures. B. The isogenic speE mutant was transformed with either pspeG(pCA24n-speG) or the VC (pCA24n) and cell growth and β -galactosidase activity were assayed as described for 3A. The values represent average promoter activity with standard deviations of five independent cultures. Linear regression comparison results on rprA promoter activity were significant for all experimental groups: WT/pspeG versus WT/VC ($t = -3.752$, $p = 0.000282$), WT/pspeG(Y135A) versus WT/VC ($t = -2.456$, $p = 0.015623$), and speE/pspeG versus speE/VC ($t = -3.470$, $p = 0.000745$).